```
load('lin_sys.mat', 'sys');

Ts = 0.05;                      % Sampling time (in seconds)
sys_dt = c2d(sys, Ts, 'zoh');  % Discrete-time state-space model

Q = diag([2, 2, 20, 1, 1, 1, 2, 2, 0.1, 0.1, 0.1, 0.1])
R = eye(4)

[K, S, e] = dlqr(sys_dt.A, sys_dt.B, Q, R, zeros(12, 4))
```

```
Q =

  Columns 1 through 7

    2.0000         0         0         0         0         0         0
         0    2.0000         0         0         0         0         0
         0         0   20.0000         0         0         0         0
         0         0         0    1.0000         0         0         0
         0         0         0         0    1.0000         0         0
         0         0         0         0         0    1.0000         0
         0         0         0         0         0         0    2.0000
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0

  Columns 8 through 12

         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
    2.0000         0         0         0         0
         0    0.1000         0         0         0
         0         0    0.1000         0         0
         0         0         0    0.1000         0
         0         0         0         0    0.1000


R =

     1     0     0     0
     0     1     0     0
     0     0     1     0
     0     0     0     1
```

```
K =

  Columns 1 through 7

    0.0000   -0.8537    2.1997    0.0000   -0.9491    2.7125    5.8786
    0.8537   -0.0000    2.1997    0.9491   -0.0000    2.7125    0.0000
   -0.0000    0.8537    2.1997   -0.0000    0.9491    2.7125   -5.8786
   -0.8537   -0.0000    2.1997   -0.9491   -0.0000    2.7125    0.0000

  Columns 8 through 12

    0.0000    0.1567    1.9515    0.0000    0.8989
    5.8786   -0.1567    0.0000    1.9515   -0.8989
   -0.0000    0.1567   -1.9515   -0.0000    0.8989
   -5.8786   -0.1567    0.0000   -1.9515   -0.8989


S =

  Columns 1 through 7

   58.5089    0.0000   -0.0000   26.9622    0.0000   -0.0000   -0.0000
    0.0000   58.5089    0.0000    0.0000   26.9622    0.0000  -84.6464
   -0.0000    0.0000  872.1111   -0.0000    0.0000  745.3084   -0.0000
   26.9622    0.0000   -0.0000   25.9949   -0.0000   -0.0000   -0.0000
    0.0000   26.9622    0.0000   -0.0000   25.9949    0.0000  -87.0698
   -0.0000    0.0000  745.3084   -0.0000    0.0000  912.4818   -0.0000
   -0.0000  -84.6464   -0.0000   -0.0000  -87.0698   -0.0000  415.8065
   84.6464   -0.0000   -0.0000   87.0698   -0.0000   -0.0000    0.0000
    0.0000   -0.0000   -0.0000    0.0000    0.0000   -0.0000   -0.0000
   -0.0000  -13.3903   -0.0000   -0.0000  -14.7293   -0.0000   88.3721
   13.3903   -0.0000   -0.0000   14.7293   -0.0000   -0.0000    0.0000
    0.0000    0.0000    0.0000   -0.0000    0.0000    0.0000   -0.0000

  Columns 8 through 12

   84.6464    0.0000   -0.0000   13.3903    0.0000
   -0.0000   -0.0000  -13.3903   -0.0000    0.0000
   -0.0000   -0.0000   -0.0000   -0.0000    0.0000
   87.0698    0.0000   -0.0000   14.7293   -0.0000
   -0.0000    0.0000  -14.7293   -0.0000    0.0000
   -0.0000   -0.0000   -0.0000   -0.0000    0.0000
    0.0000   -0.0000   88.3721    0.0000   -0.0000
  415.8065    0.0000    0.0000   88.3721   -0.0000
    0.0000   11.4734   -0.0000    0.0000   31.6228
    0.0000   -0.0000   28.2695    0.0000   -0.0000
   88.3721    0.0000    0.0000   28.2695   -0.0000
   -0.0000   31.6228   -0.0000   -0.0000  180.6698


e =

   0.9910 + 0.0087i
   0.9910 - 0.0087i
```

```
0.9713 + 0.0218i
0.9713 - 0.0218i
0.9248 + 0.1133i
0.9248 - 0.1133i
0.9248 + 0.1133i
0.9248 - 0.1133i
0.9299 + 0.0000i
0.9299 + 0.0000i
0.8805 + 0.0000i
0.8805 + 0.0000i
```

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